

Unequal Job Opportunities and the Measurement of Welfare Inequality

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France 2016 · EU-SILC priced through EUROMOD

Two people, the same job, different reasons

Person A

40 hours

same occupation

bottom wage quintile

chose it from a wide set

Person B

40 hours

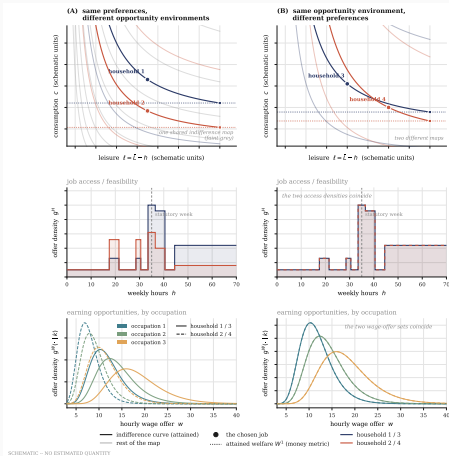
same occupation

bottom wage quintile

had almost nothing else

Whatever the choice set cannot express,
the taste parameter absorbs.

What the model has to separate



Schematic. Stylised parameters; no estimated quantity on any axis.

Why random opportunities, not a common choice set

	common choice set	random opportunities
what varies across households	the budget only	the whole offer density
unexplained variation goes to	preferences	job access, earning opportunities, preferences
welfare comparison available	levels at a common set	levels at own set
what is identified	tastes	tastes <i>and</i> availability

Remove the occupation term and the effect does not vanish —
it relocates.

tertiary-education loading moves by six standard errors

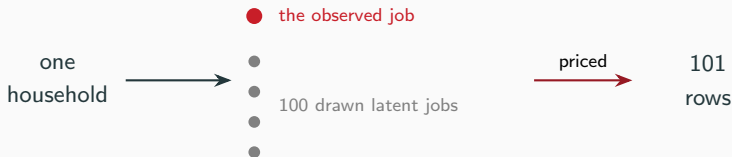
What the opportunity distribution is made of



estimated jointly with Box–Cox **preferences** in a single likelihood

The hours factor carries no household-specific coordinate.

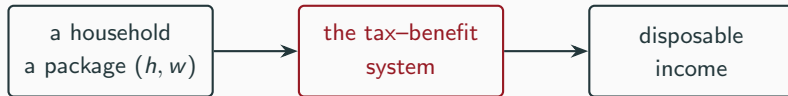
Data and the sampled-job estimator



EU-SILC / SRCV 2016 · 1,555 households · 101 alternatives each · 157,055
priced rows

conditional logit with the sampling-of-alternatives correction

Pricing: what a job is worth



Consumption is computed, not surveyed.

The mapping is deterministic: the same household, hours and wage always return the same euro figure.

Nothing in the estimation simulates a tax-benefit outcome.

Identification: what is pinned, what stays conditional

what the data pin	what stays conditional
the four availability factors	access <i>versus</i> capability
the preference block	persistent household heterogeneity
the statutory-week peak	

The decomposition is structural and model-conditional. It is not causal.

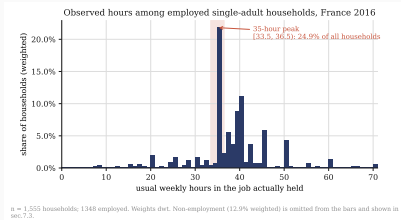
Three heterogeneity extensions

extension	what defeats recovery
random leisure intercept	no leverage
working-alternative frailty	leverage, but no second contrast
wage-residual dependence	no leverage

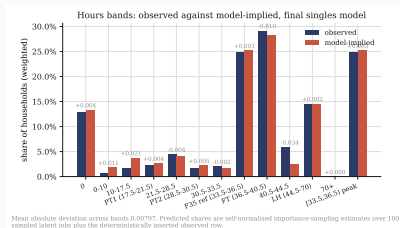
Depth on an axis is not evidence; only depth on a profile is.

axis 13.56 $\longrightarrow$ profile 0.6996 against 2.706

Hours, and the statutory week



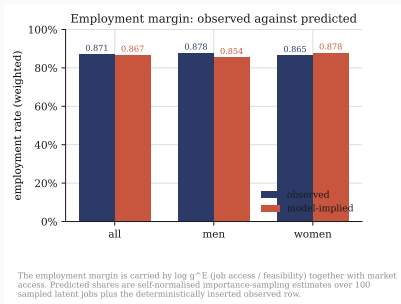
observed hours



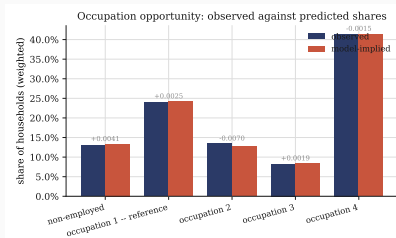
observed against model-implied

statutory band 24.85% observed · 25.20% model-implied · mean absolute error 0.0083 over 12 bins

Employment, occupation, and wages



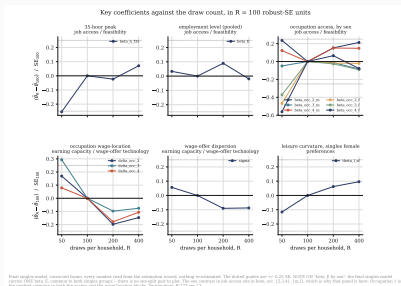
employment



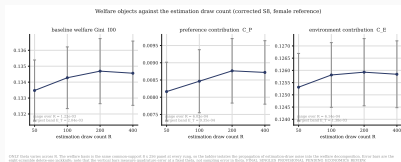
occupation shares

employment 0.8708 / 0.8668 · occupation largest deviation 0.0070 · median wage fit -0.86 men, +0.34 women (€/h)

Does the answer depend on the number of drawn jobs?



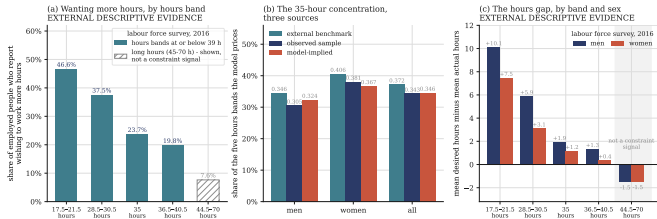
key coordinates, in SE units



the headline contributions

No. Stable from 50 to 400 drawn alternatives.

External validation



All three panels are external evidence about hours. No moment from this source enters the model's objective, and none of it identifies a parameter. The population is all persons aged 20-60 with a reported usual hours figure (43,767 records); the estimation sample is single-adult households, so composition differs by construction and no gap between them is a prediction error. Panel (a). Sexes pooled, because the wish-to-work more question was returned pooled. Panels (a) and (c) are EXTERNAL DESCRIPTIVE EVIDENCE ONLY: the structural model has no desired hours or underemployment outcome, so these moments are not compared with a model prediction, and identify nothing in the model. No model quantity appears on either axis of either panel, and no rank statistic against an estimated offer weight is computed or shown. Panel (a) is coded from the survey's wish-to-work-more variable as 1 = No, 2 = Yes per the published variable codebook (version of 8 July 2021); the delivered export's own field name had the map the other way round and is superseded. Across the five bands 20.3% of employed people wish to work more hours. The long hours band is drawn but marked, because a band can be thin as OFFERS while the people already working 45-70 hours are the least likely to want more. Panel (b) is the validation panel and does not use the wish-to-work-more variable. All three sources are put on the share of the five bands the model prices, the only base on which they are commensurable. Panel (c) is mean desired hours minus mean ACTUAL hours within the same band, both weighted. The gap is positive on every band at or below 39 hours for both sexes and turns negative only at long hours. It is larger for men than for women in all four short-and-medium bands.

statutory band 37.2% external · 34.3% observed · 34.6% model-implied

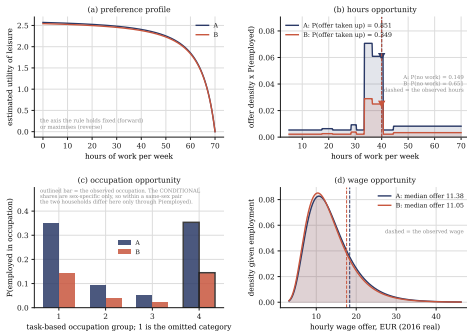
The welfare measure, and two real households

Same estimated preferences, same observed job, most different estimated opportunities

A: male, 35-39, high education, no children, intermediate

B: male, 40-44, low education, no children, rural

matched on the observed job: both employed, occupation 4, hours band "H", wage quintile 4. d_pref = 0.021 (admissible median 0.783). d_opp = 0.502 (admissible median 0.209).

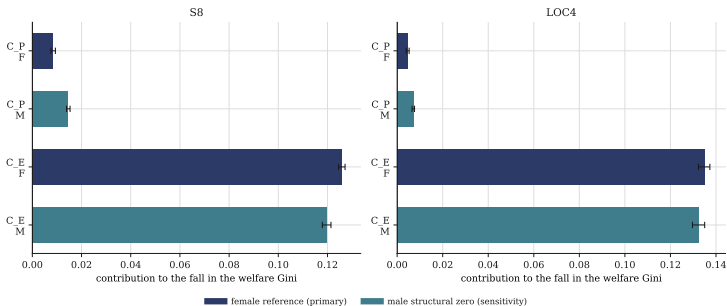


Final singles model, corrected frame: the pair is the argmax of the stated rule over 19,116 admissible employed pairs, not a hand pick. Panels (b)-(d) are the model's OFFER distribution, not the choice distribution. In the final singles model the hours band rules carries no household-specific coordinate, so the two hours curves differ only by the scalar P(employed); the model lets opportunity differ across households in the participation margin, the sex-specific occupation shares and the wage offer location, and nowhere else. Terminology: B.227 sec.12 - job access / feasibility, earning capacity / wage-offer technology, preferences. No household identifier is used.

same preference profile (4.61 and 4.56) · employment probability **0.851** against **0.349**

The headline

Preference and environment contributions, both reference conventions (raw welfare)

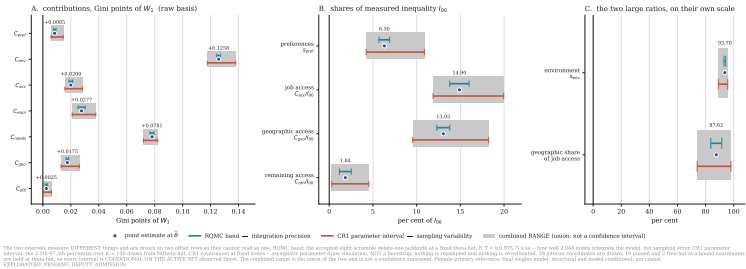


Owen value for the partition $\{ \{ \text{preferences} \}, \{ \text{job access, earning opportunities, endowments-needs} \} \}$ on $v(S) = I(\text{baseline}) - I(S)$. Bars are SIGNED and are never renormalised. Error bars are the eight-scramble delete-one jackknife at $t = 2.3646$. F = female reference (primary), M = male structural zero (sensitivity); the two are NEVER averaged.
FINAL_SINGLES_PROVISIONAL_PENDING_ECONOMICS_REVIEW

The environment $93.70 \pm 0.61\%$ · Preferences
 $6.30 \pm 0.61\%$

The headline

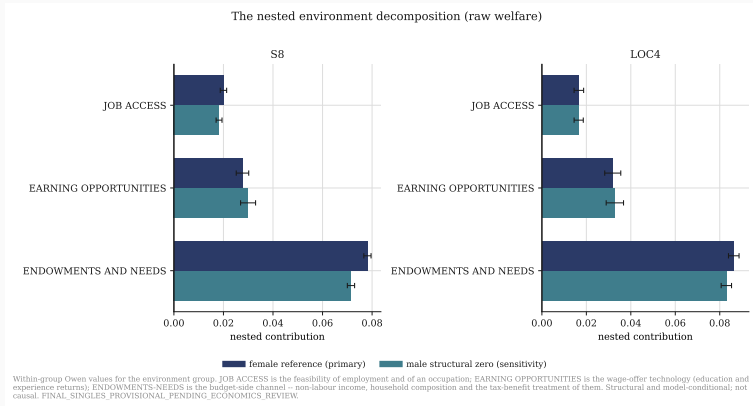
The headline decomposition under two kinds of uncertainty, kept apart



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parameter interval: preferences [4.22, 10.89] the environment [89.11, 95.78]

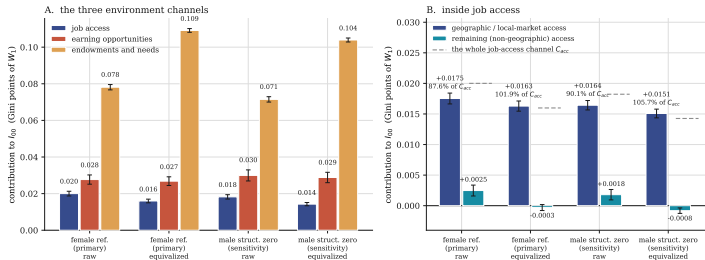
Inside the environment



Household endowments and needs $58.18 \pm 1.47\%$ · Earning opportunities
20.62 $\pm 1.63\%$ · Job access 14.90 $\pm 1.10\%$
market side together: $35.51 \pm 0.95\%$

Inside job access: it is almost all geography

The nested geographic split of the job-access channel (final singles model, money-metric W_1)

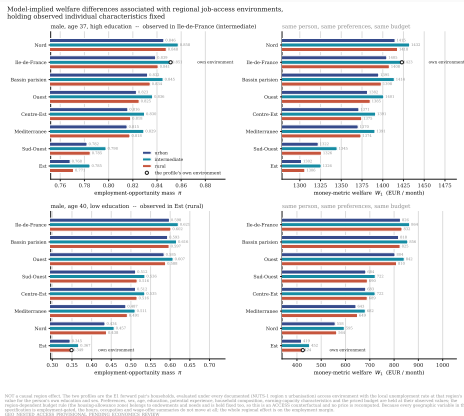


Owen / level-structure contributions on the money-metric welfare measure W_1 , with the eight-scramble delete-one jackknife band $E.T = (10.975, 7) \times \text{se}$. Geographic access = the NUTS-1 region indicators, the urbanisation indicators and the local unemployment rate; remaining access = the occupation-availability conditional and the survey-year indicators. $C_{geo} + C_{oth} = C_{acc}$ exactly, by construction of the level-structure value and asserted at machine precision. Structural and model-conditional, not causal. GEO_NESTED_ACCESS_PROVISIONAL_PENDING_ECONOMICS_REVIEW

geographic access $13.05 \pm 0.74\%$ raw, $9.87 \pm 0.55\%$ equivalized — 87.62% and 101.92% of **job access** [74, 98]

This is not a finding that region drives welfare inequality.

Inside job access: it is almost all geography

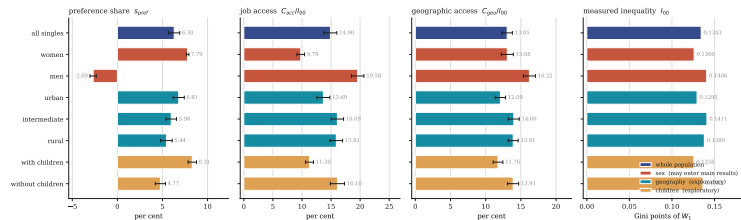


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Job access does more of the work among men

The decomposition re-aggregated within each subgroup, on population-level operators



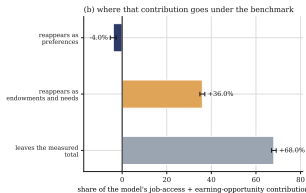
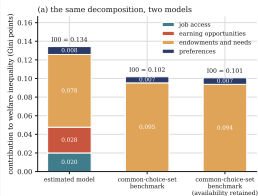
THE OPERATORS ARE POPULATION-LEVEL. Every reference object – the substituted argument vector, the reference preference block, the frozen occupation joint with its dot population SEX shares, and the channel D modelled profile with its equivalence scale – is computed once over the whole singles population and is COMMON to every group. A subgroup number therefore answers how much of THAT group's own money-welfare inequality is removed by moving its households to the COMMON population reference; it does not answer what a within-group reference would give, which is a different estimate and is not computed. The Gini is incorporated inside the group, hence are the eight-separable double-one percentile (integration precision) computed inside the group, and carry no parameter uncertainty. Exhaustiveness and the three accounting identities are re-checked in every group: 32/32 cells PASS. Female: primary reference; raw basis. EXPLORATORY, PENDING, DEPUTY, ADMISSION

job access: $19.58 \pm 1.08\%$ of men's measured inequality against $9.79 \pm 0.65\%$ of women's

Men's preferences share: -2.69% and $+2.78\%$ across the two reference conventions —
a sign change. No sign statement is licensed.

What a conventional model would have found

Welfare-inequality decomposition: estimated model against the common-choice-set benchmark



Single-adult households, France 2010. Money-metric: welfare level W1, female reference, raw basis. Own values on the grouped game: bands are the eight-scramble jackknife at $t = 2.364624251$. FINAL_RUM_BENCHMARK_PROVISIONAL_PENDING_ECONOMICS_REVIEW

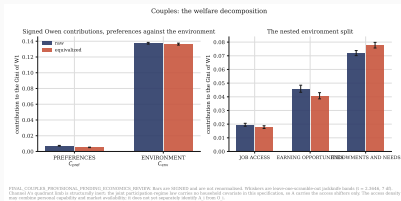
preferences share 6.4% against 6.3% · measured inequality -24.2% raw · sex
leisure gap +0.428 → -1.991

What is fragile, and what is not

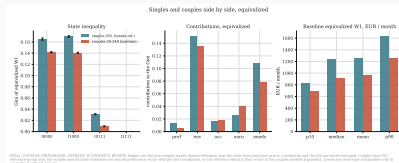
perturbation	preferences share	the environment side
reference-preference convention	6.30 $\rightarrow$ 10.89%	$< 3.5\%$
a bound on a preference curvature	$\approx 13\%$ move	$< 3.5\%$
pinning the couples male leisure block	up to 73% move	$< 3.5\%$

No sign and no ordering changes anywhere. Market-side total 35.51% against 35.88% across references.

Couples



the couples decomposition



singles and couples, equivalized

2,275 couples · same nested order, same the environment dominance

The couples preferences share is not structurally robust. Singles remain the quantitative headline.

Limits, and what would identify more

limit	what would change it
access versus capability	external job-offer or access-rate moments
persistent heterogeneity	repeated choices per household
no desired-hours target	external desired-hours moments
one country, one year, one system	re-estimation; no portability claimed

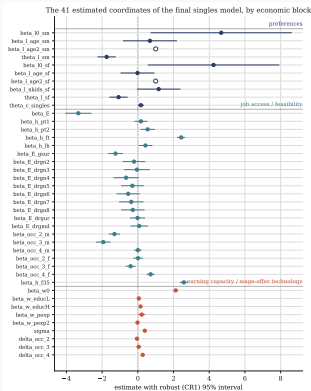
Multi-adult households are excluded. Welfare magnitudes are provisional pending one bounded economics review.

- **The method.** A household-specific opportunity distribution estimated jointly with preferences, every job priced through the real tax–benefit system.

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- **The method.** A household-specific opportunity distribution estimated jointly with **preferences**, every job priced through the real tax–benefit system.
- **The finding.** The environment carries 93.70% of measured money-metric welfare inequality; inside it the budget side carries more than the market side.
- **The margin.** The **preferences** share is the fragile quantity: 6.30% to 10.89% across the two normative conventions.

B1 — The estimated coefficients

[illegible]

41 estimated structural parameters, robust 95% intervals

B2 — Exhaustiveness: how the decomposition was made to close

defect	what fixed it
a missing budget channel	added as a fourth player from its own marginals
asymmetric money-metric inversion	reference side moves with the coalition
an incomplete male reference block	structural zero on the child leisure term

Exhaustiveness here is a tested property, pre-stated before any cell was read.

the fully common state is 0.000000 — numerically zero

B3 — Equivalization, and why the scale has to move

	raw	equivalized
preferences	6.30%	7.98%
the environment	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing each household's own scale leaves a residual of 0.071799 — precisely the weighted Gini of one over the scale.

B4 — Why the hours-wish evidence is context, not a moment

ground	what it shows
definitional	the candidate's optimum tracks pay: 98.1% pay more, median 2.58×
identification	the hours block moves the moment by exactly 0
numerical	24.3% against 20.3%; the profile does not match

Steelman (own $\pm 5\%$ wage neighbourhood): 19.4% — and the hours block is exactly as inert.

B5 — Where the welfare family comes from

Compensation

unequal productivity should not
become unequal well-being

Responsibility

preference-based choices should not
be compensated

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet, SCW 2005]

$W^1(z, R, A; \mathbf{y}^w)$ — equal pay on the feasible set;
the measure this paper takes to the data

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).